

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, wordclass changes, and semantic shifts.	Q4

1. Search Engine Indexing System

Scenario

A search engine processes the words **"analyzing"**, **"analysis"**, and **"analytical"**. To improve search accuracy, the system groups related words under one common representation using morphological analysis.

Step 1: Morphological Decomposition

Word	Root	Affix	Word Class
analyzing	analyze	-ing	Verb (Present Participle)
analysis	analyze	-sis	Noun
analytical	analyze	-ical	Adjective

Step 2: Inflectional vs Derivational

- **analyzing** → **Inflectional**
 - "-ing" indicates an action in progress.
 - Word class remains a verb.
- **analysis** → **Derivational**
 - "-sis" converts the verb into a noun.
- **analytical** → **Derivational**
 - "-ical" converts the base into an adjective.

Step 3: Normalization

All variants are mapped to the common base form:

analyzing

analysis

analytical



analyze

Final Morphological Structure

Input Word	Morphological Structure	Classification	Normalized Form
analyzing	analyze + ing	Inflectional	analyze
analysis	analyze + sis	Derivational	analyze
analytical	analyze + ical	Derivational	analyze

Morphological Processing Pipeline

Input Words

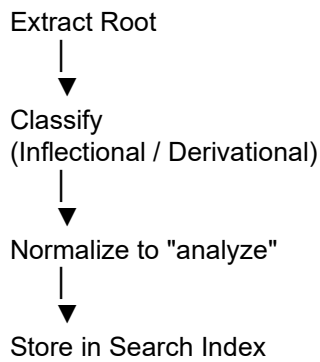


Tokenization



Identify Prefix/Suffix





2. AI-Based Sentiment Analysis

Scenario

The system receives the words:

- disagree
- agreement
- agreeable

It must identify the common base word for consistent sentiment interpretation.

Morphological Parsing

Word 1: disagree

dis + agree

Prefix = dis-

Root = agree

Word 2: agreement

agree + ment

Suffix = -ment

Forms a noun.

Word 3: agreeable

agree + able

Suffix = -able

Forms an adjective.

Classification

Word	Prefix	Root	Suffix	Type
disagree	dis-	agree	—	Derivational
agreement	—	agree	-ment	Derivational
agreeable	—	agree	-able	Derivational

Meaning Change

- **agree** → to have the same opinion
- **disagree** → opposite meaning
- **agreement** → state of agreeing
- **agreeable** → pleasant or willing to agree

Normalized Representation

disagree

agreement
agreeable
↓
agree

Final Output

Word	Morphological Breakdown	Classification	Root
disagree	dis + agree	Derivational	agree
agreement	agree + ment	Derivational	agree
agreeable	agree + able	Derivational	agree

3. News Aggregation System

Scenario

Input Words

- govern
- government
- governance

The system standardizes these words for topic modeling.

Morphological Decomposition

govern

govern

Root word

government

govern + ment

Creates a noun.

governance

govern + ance

Creates a noun.

Derivational Layers

Word	Root	Suffix	Type
govern	govern	—	Root
government	govern	-ment	Derivational
governance	govern	-ance	Derivational

Normalization

govern

government

governance

↓

govern

Final Output

Input	Morphological Structure	Normalized Form
govern	govern	govern
government	govern + ment	govern
governance	govern + ance	govern

Processing Flow

Input



Morphological Analysis



Extract Root



Remove Derivational Suffix



Normalize



Store "govern"

4. Document Classification System

Scenario

Input Words

- activate
- activation
- reactivation

The system groups related words by identifying prefixes, suffixes, and root forms.

Morphological Parsing

activate

activate

Root word (verb)

activation

activate + ion

Verb → Noun

reactivation

re + activate + ion

Prefix = re-

Suffix = -ion

Morphological Breakdown

Word	Prefix	Root	Suffix	Word Class
activate	—	activate	—	Verb
activation	—	activate	-ion	Noun
reactivation	re-	activate	-ion	Noun

Meaning Changes

- **activate** → make active
- **activation** → process of making active
- **reactivation** → making active again

Classification

Word	Type
activate	Base Form
activation	Derivational
reactivation	Derivational

Unified Representation

activate

activation

reactivation

↓

activate

Final Output

Word	Morphological Structure	Normalized Root
activate	activate	activate
activation	activate + ion	activate
reactivation	re + activate + ion	activate

5. Search Optimization Module

Scenario

Input Words

- create
- creates
- creating

The system removes only inflectional changes for consistent retrieval.

Morphological Decomposition

create

create

Base word

creates

create + s

Third-person singular verb

creating

create + ing

Present participle

Classification

Word	Root	Suffix	Type
create	create	—	Base Form
creates	create	-s	Inflectional
creating	create	-ing	Inflectional

Normalization

create
creates
creating

↓
create

Final Output

Input Word	Morphological Breakdown	Classification	Normalized Form
create	create	Base Form	create
creates	create + s	Inflectional	create
creating	create + ing	Inflectional	create

Morphology-Based Normalization Flow

Input Words



Detect Suffix



Remove Inflectional Ending



Extract Root



Normalize to "create"



Store in Search Index

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Rubrics for Calculation
